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Multilayer high precision PCB's with impedance control

JLCPCB can produce High-precision multilayer board with capabilities listed in below table.

Min. Trace width/Spacing	Min. Via	Min. BGA
3.5mil	0.2mm	0.25mm

It is very convenient for customers to conduct an impedance matching design, according to JLCPCB's laminated structure and related parameters. The PCB will be strictly produced in accordance with the following stackup.

Controlled Impedance PCB Parameters and Stackup

1.Prepreg dielectric constant:

Prepreg type	Dielectric constant
7628	4.4
3313	4.1
1080	3.91
2116	4.16

2.Solder mask Parameters

Coating Above Substrate C1	Coating Above Trace C2	Coating Between Traces C3	Coating Dielectric CEr
1.2mil	0.6mil	1.2mil	3.8

3. Core dielectric constant

Core dielectric constant
4.6

For your convenience, we have designed an Impedance Calculator to help you calculate the impedance and the trace width you require.

4-Layer Impedance Control Stackup

Thickness: 0.8mm | 1.0mm | 1.2mm | 1.6mm | 2.0mm

Outer Copper Weight: 1oz | 2oz

inner Copper Weight: 0.5oz | 1oz | 2oz

1) No requirement Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	7628*1	0.21040mm	
Inner Layer L2	Copper	0.0152mm	1.1mm H/HOZ with copper
Core	Core	1.065mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.21040mm	
Bottom Layer	Copper	0.035mm	

2) JLC04161H-7628 Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	7628*1	0.21040mm	
Inner Layer L2	Copper	0.0152mm	1.1mm H/HOZ with copper
Core	Core	1.065mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.21040mm	
Bottom Layer	Copper	0.035mm	

3) JLC04161H-3313 Stackup

Layer	Material Type	Thickness	

Top Layer	Copper	0.035mm	
Prepreg	3313*1	0.09940mm	
Inner Layer L2	Copper	0.0152mm	1.3mm H/HOZ with copper
Core	Core	1.265mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	3313*1	0.09940mm	
Bottom Layer	Copper	0.035mm	

4) JLC04161H-1080 Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	1080*1	0.07640mm	
Inner Layer L2	Copper	0.0152mm	1.3mm H/HOZ with copper
Core	Core	1.265mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	1080*1	0.07640mm	
Bottom Layer	Copper	0.035mm	

5) JLC04161H-7628A Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	7628*1	0.218mm	
Prepreg	1080*1	0.07640mm	
Inner Layer L2	Copper	0.0152mm	0.9mm H/HOZ with copper
Core	Core	0.865mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	1080*1	0.07640mm	
Prepreg	7628*1	0.218mm	
Bottom Layer	Copper	0.035mm	

6) JLC04161H-3313A Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	3313*1	0.107mm	
Prepreg	3313*1	0.09940mm	
Inner Layer L2	Copper	0.0152mm	1.1mm H/HOZ with copper
Core	Core	1.065mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	3313*1	0.09940mm	
Prepreg	3313*1	0.107mm	
Bottom Layer	Copper	0.035mm	

7) JLC04161H-1080A Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	1080*1	0.084mm	
Prepreg	1080*1	0.07640mm	
Inner Layer L2	Copper	0.0152mm	1.1mm H/HOZ with copper
Core	Core	1.065mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	1080*1	0.07640mm	
Prepreg	1080*1	0.084mm	
Bottom Layer	Copper	0.035mm	

8) JLC04161H-7628B Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	7628*1	0.218mm	
Prepreg	7628*1	0.218mm	
Prepreg	2116*1	0.11640mm	
Inner Layer L2	Copper	0.0152mm	

Core	Core	0.4mm	0.4mm H/HOZ without copper
Inner Layer L3	Copper	0.0152mm	
Prepreg	2116*1	0.11640mm	
Prepreg	7628*1	0.218mm	
Prepreg	7628*1	0.218mm	
Bottom Layer	Copper	0.035mm	

9) JLC04161H-2116A Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	2116*1	0.124mm	
Prepreg	7628*1	0.218mm	
Prepreg	2116*1	0.11640mm	
Inner Layer L2	Copper	0.0152mm	0.6mm H/HOZ without copper
Core	Core	0.6mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	2116*1	0.11640mm	
Prepreg	7628*1	0.218mm	
Prepreg	2116*1	0.124mm	
Bottom Layer	Copper	0.035mm	

10) JLC04161H-2116B Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	2116*1	0.124mm	
Prepreg	7628*1	0.21040mm	
Inner Layer L2	Copper	0.0152mm	0.9mm H/HOZ with copper
Core	Core	0.865mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.21040mm	
Prepreg	2116*1	0.124mm	
Bottom Layer	Copper	0.035mm	

11) JLC04161H-2116C Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	2116*1	0.124mm	
Prepreg	1080*1	0.07640mm	
Inner Layer L2	Copper	0.0152mm	
Core	Core	1.065mm	1.1mm H/HOZ with copper
Inner Layer L3	Copper	0.0152mm	
Prepreg	1080*1	0.07640mm	
Prepreg	2116*1	0.124mm	
Bottom Layer	Copper	0.035mm	

12) JLC04161H-7628G Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	7628*1	0.218mm	
Prepreg	7628*1	0.218mm	
Prepreg	1080*1	0.07640mm	
Inner Layer L2	Copper	0.0152mm	0.5mm H/HOZ without copper
Core	Core	0.5mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	1080*1	0.07640mm	
Prepreg	7628*1	0.218mm	
Prepreg	7628*1	0.218mm	
Bottom Layer	Copper	0.035mm	

13) JLC04161H-2116 Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	2116*1	0.11640mm	
Inner Layer L2	Copper	0.0152mm	1.3mm H/HOZ with copper
Core	Core	1.265mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	2116*1	0.11640mm	
Bottom Layer	Copper	0.035mm	

14) JLC04161H-7628E Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	7628*1	0.218mm	
Prepreg	7628*1	0.21040mm	
Inner Layer L2	Copper	0.0152mm	0.6mm H/HOZ without copper
Core	Core	0.6mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.21040mm	
Prepreg	7628*1	0.218mm	
Bottom Layer	Copper	0.035mm	

15) JLC04161H-2116D Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	2116*1	0.124mm	
Prepreg	7628*1	0.21040mm	
Inner Layer L2	Copper	0.0152mm	0.7mm H/HOZ without copper
Core	Core	0.7mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.21040mm	
Prepreg	2116*1	0.124mm	
Bottom Layer	Copper	0.035mm	

16) JLC04161H-7628F Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	7628*1	0.218mm	
Prepreg	7628*1	0.218mm	
Prepreg	7628*1	0.21040mm	
Inner Layer L2	Copper	0.0152mm	0.25mm H/HOZ without copper
Core	Core	0.25mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.21040mm	
Prepreg	7628*1	0.218mm	
Prepreg	7628*1	0.218mm	
Bottom Layer	Copper	0.035mm	

17) JLC04161H-2116E Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	2116*1	0.124mm	
Prepreg	2116*1	0.11640mm	
Inner Layer L2	Copper	0.0152mm	0.9mm H/HOZ with copper
Core	Core	0.865mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	2116*1	0.11640mm	
Prepreg	2116*1	0.124mm	
Bottom Layer	Copper	0.035mm	

18) JLC04161H-7628C Stackup

Layer	Material Type	Thickness	

Top Layer	Copper	0.035mm	
Prepreg	7628*1	0.218mm	
Prepreg	7628*1	0.218mm	
Prepreg	7628*1	0.21040mm	
Inner Layer L2	Copper	0.0152mm	0.15mm H/HOZ without copper
Core	Core	0.15mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.21040mm	
Prepreg	7628*1	0.218mm	
Prepreg	7628*1	0.218mm	
Bottom Layer	Copper	0.035mm	

6-Layer Impedance Control Stackup

Thickness: 1.2mm | 1.6mm | 2.0mm

Outer Copper Weight: 1oz | 2oz

inner Copper Weight: 0.5oz | 1oz | 2oz

1) No requirement Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	3313*1	0.09940mm	
Inner Layer L2	Copper	0.0152mm	0.55mm H/Hwithout copper
Core	Core	0.55mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	2116*1	0.11640mm	
Inner Layer L4	Copper	0.0152mm	0.55mm H/Hwithout copper
Core	Core	0.55mm	
Inner Layer L5	Copper	0.0152mm	
Prepreg	3313*1	0.09940mm	
Bottom Layer	Copper	0.035mm	

2) JLC06161H-3313 Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	3313*1	0.09940mm	
Inner Layer L2	Copper	0.0152mm	0.55mm H/Hwithout copper
Core	Core	0.55mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	2116*1	0.1088mm	
Inner Layer L4	Copper	0.0152mm	0.55mm H/Hwithout copper
Core	Core	0.55mm	
Inner Layer L5	Copper	0.0152mm	
Prepreg	3313*1	0.09940mm	
Bottom Layer	Copper	0.035mm	

3) JLC06161H-7628 Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	7628*1	0.21040mm	
Inner Layer L2	Copper	0.0152mm	0.4mm H/HOZ without copper
Core	Core	0.4mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.2028mm	
Inner Layer L4	Copper	0.0152mm	0.4mm H/HOZ without copper
Core	Core	0.4mm	
Inner Layer L5	Copper	0.0152mm	
Prepreg	7628*1	0.21040mm	
Bottom Layer	Copper	0.035mm	

4) JLC06161H-7628D Stackup

Layer	Material Type	Thickness	

Top Layer	Copper	0.035mm	
Prepreg	7628*1	0.216mm	
Prepreg	7628*1	0.20840mm	
Inner Layer L2	Copper	0.0152mm	0.2mm H/HOZ without copper
Core	Core	0.2mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.20080mm	
Inner Layer L4	Copper	0.0152mm	0.2mm H/HOZ without copper
Core	Core	0.2mm	
Inner Layer L5	Copper	0.0152mm	
Prepreg	7628*1	0.20840mm	
Prepreg	7628*1	0.216mm	
Bottom Layer	Copper	0.035mm	

5) JLC06161H-1080 Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	1080*1	0.07640mm	
Inner Layer L2	Copper	0.0152mm	
Core	Core	0.55mm	0.55mm H/Hwithout copper
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.21040mm	
Inner Layer L4	Copper	0.0152mm	
Core	Core	0.55mm	
Inner Layer L5	Copper	0.0152mm	
Prepreg	1080*1	0.07640mm	
Bottom Layer	Copper	0.035mm	

6) JLC06161H-2116A Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	2116*1	0.11640mm	
Inner Layer L2	Copper	0.0152mm	0.13mm H/HOZ without copper
Core	Core	0.13mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	2116*1	0.11640mm	
Core	Core	0.7mm	0.7 H/HOZ without copper
Prepreg	2116*1	0.11640mm	
Inner Layer L4	Copper	0.0152mm	0.13mm H/HOZ without copper
Core	Core	0.13mm	
Inner Layer L5	Copper	0.0152mm	
Prepreg	2116*1	0.11640mm	
Bottom Layer	Copper	0.035mm	

7) JLC06161H-1080A Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	1080*1	0.07640mm	
Inner Layer L2	Copper	0.0152mm	0.6mm H/HOZ without copper
Core	Core	0.6mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	3313*1	0.09940mm	
Inner Layer L4	Copper	0.0152mm	0.6mm H/HOZ without copper
Core	Core	0.6mm	
Inner Layer L5	Copper	0.0152mm	
Prepreg	1080*1	0.07640mm	
Bottom Layer	Copper	0.035mm	

8) JLC06161H-3313C Stackup

Layer	Material Type	Thickness	

Top Layer	Copper	0.035mm	
Prepreg	3313*1	0.107mm	
Prepreg	1080*1	0.07640mm	
Inner Layer L2	Copper	0.0152mm	0.4mm H/HOZ without copper
Core	Core	0.4mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	1080*1	0.07640mm	
Prepreg	1080*1	0.07640mm	
Inner Layer L4	Copper	0.0152mm	0.4mm H/HOZ without copper
Core	Core	0.4mm	
Inner Layer L5	Copper	0.0152mm	
Prepreg	1080*1	0.07840mm	
Prepreg	3313*1	0.107mm	
Bottom Layer	Copper	0.035mm	

9) JLC06161H-1080B Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	1080*1	0.07640mm	
Inner Layer L2	Copper	0.0152mm	0.10mm H/HOZ without copper
Core	Core	0.1mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.21040mm	
Core	Core	0.7mm	0.7 H/HOZ without copper
Prepreg	7628*1	0.21040mm	
Inner Layer L4	Copper	0.0152mm	0.10mm H/HOZ without copper
Core	Core	0.1mm	
Inner Layer L5	Copper	0.0152mm	
Prepreg	1080*1	0.07640mm	
Bottom Layer	Copper	0.035mm	

10) JLC06161H-3313E Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	3313*1	0.09940mm	
Inner Layer L2	Copper	0.0152mm	0.10mm H/HOZ without copper
Core	Core	0.1mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.21040mm	
Inner Layer L4	Copper	0.0152mm	0.10mm H/HOZ without copper
Core	Core	0.1mm	
Inner Layer L5	Copper	0.0152mm	
Prepreg	3313*1	0.09940mm	
Bottom Layer	Copper	0.035mm	

11) JLC06161H-2116B Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	2116*1	0.11640mm	
Inner Layer L2	Copper	0.0152mm	0.5mm H/HOZ without copper
Core	Core	0.5mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	1080*1	0.07640mm	
Prepreg	1080*1	0.07640mm	
Inner Layer L4	Copper	0.0152mm	0.5mm H/HOZ without copper
Core	Core	0.5mm	
Inner Layer L5	Copper	0.0152mm	

Prepreg	2116*1	0.11640mm	
Bottom Layer	Copper	0.035mm	

12) JLC06161H-2116 Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	2116*1	0.127mm	
Prepreg	2313*1	0.09640mm	
Inner Layer L2	Copper	0.0152mm	0.30mm H/HOZ without copper
Core	Core	0.3mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.20840mm	
Prepreg	7628*1	0.20840mm	
Inner Layer L4	Copper	0.0152mm	0.30mm H/HOZ without copper
Core	Core	0.3mm	
Inner Layer L5	Copper	0.0152mm	
Prepreg	2313*1	0.09640mm	
Prepreg	2116*1	0.127mm	
Bottom Layer	Copper	0.035mm	

13) JLC06161H-7628B Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	7628*1	0.218mm	
Prepreg	1080*1	0.07640mm	
Inner Layer L2	Copper	0.0152mm	0.35mm H/HOZ without copper
Core	Core	0.35mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	1080*1	0.07640mm	
Prepreg	1080*1	0.07640mm	
Inner Layer L4	Copper	0.0152mm	0.35mm H/HOZ without copper
Core	Core	0.35mm	
Inner Layer L5	Copper	0.0152mm	
Prepreg	1080*1	0.07640mm	
Prepreg	7628*1	0.218mm	
Bottom Layer	Copper	0.035mm	

14) JLC06161H-3313D Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	3313*1	0.107mm	
Prepreg	1080*1	0.07640mm	
Inner Layer L2	Copper	0.0152mm	0.25mm H/HOZ without copper
Core	Core	0.25mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.21040mm	
Prepreg	2116*1	0.124mm	
Prepreg	7628*1	0.21040mm	
Inner Layer L4	Copper	0.0152mm	0.25mm H/HOZ without copper
Core	Core	0.25mm	
Inner Layer L5	Copper	0.0152mm	
Prepreg	1080*1	0.07640mm	
Prepreg	3313*1	0.107mm	
Bottom Layer	Copper	0.035mm	

15) JLC06161H-2116C Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	2116*1	0.127mm	
Prepreg	2116*1	0.11940mm	

Inner Layer L2	Copper	0.0152mm	
Core	Core	0.3mm	0.30mm H/HOZ without copper
Inner Layer L3	Copper	0.0152mm	
Prepreg	2116*1	0.11940mm	
Prepreg	2116*1	0.127mm	
Prepreg	2116*1	0.11940mm	
Inner Layer L4	Copper	0.0152mm	
Core	Core	0.3mm	0.30mm H/HOZ without copper
Inner Layer L5	Copper	0.0152mm	
Prepreg	2116*1	0.11940mm	
Prepreg	2116*1	0.127mm	
Bottom Layer	Copper	0.035mm	

16) JLC06161H-7628A Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	7628*1	0.218mm	
Prepreg	7628*1	0.218mm	
Prepreg	1080*1	0.07640mm	
Inner Layer L2	Copper	0.0152mm	
Core	Core	0.1mm	0.10mm H/HOZ without copper
Inner Layer L3	Copper	0.0152mm	
Prepreg	2116*1	0.11640mm	
Prepreg	2116*1	0.11640mm	
Inner Layer L4	Copper	0.0152mm	
Core	Core	0.1mm	0.10mm H/HOZ without copper
Inner Layer L5	Copper	0.0152mm	
Prepreg	1080*1	0.07640mm	
Prepreg	7628*1	0.218mm	
Prepreg	7628*1	0.218mm	
Bottom Layer	Copper	0.035mm	

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